LUKE THOMPSON

UNREAL ENGINE 5 GAMEPLAY PROGRAMMER (C++)

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Portfolio: xylardark.github.io/Portfolio-Studio/u/admin · GitHub: github.com/XylarDark

PROFESSIONAL SUMMARY

Gameplay programmer building HomeWorld solo in Unreal Engine 5.8 and C++ since February 2026: character movement, Gameplay Ability System abilities, interaction, NPC behaviour, and designer-tunable data-driven systems, backed by C++ automation tests. Shipped Shift, a playable Phaser 3 / TypeScript game, and built game UI/UX in C# at OutFusion (2022). Self-directed remote worker in Canada (US tax-treaty country), Eastern Time.

CORE SKILLS

Languages	C++, C#, Python, TypeScript/JavaScript, SQL
Unreal Engine 5	Gameplay Ability System, custom CharacterMovementComponent modes, Enhanced Input, Blueprints, AI controllers, PCG, Automation tests
Other engines	Phaser 3, Unity
Workflow	Git/GitHub pull requests, Visual Studio, Python editor tooling, playtesting and bug verification

SELECTED PROJECTS

HomeWorld — Unreal Engine 5.8 / C++ game — Feb 2026 – present

Independent (solo) · UE 5.8, C++, Python · private repository, available on request

- Built gameplay in C++ for a third-person ARPG survival and family-sim: character, Enhanced Input, a time-of-day subsystem that gates body and spirit forms, inventory, crafting, and six GAS abilities (interact, place, heal, spirit shield and burst, protector attack) on a C++ base class made for Blueprint children.
- Wrote a custom glide movement mode by subclassing the single CharacterMovementComponent (`PhysCustom`: forward drive against a fixed sink rate, time-constant steering), plus a traversal component with sprint, trace-based mantle and vault, spirit blink with cooldown, and a fall soft-reset that probes for landing ground. Fixed a same-tick re-entry stack overflow in the glide-to-walk landing hand-off.
- Made systems designer-editable without code changes: tuning exposed as editor properties, interactions as a priority-ordered verb table (a new prop is a row, not a branch), and seeded, deterministic zone generation read from JSON zone specs. Added NPC follow and escort AI, a beast-taming state flow, and a spirit-stealth light and alert system.
- Kept stability up with 42 C++ automation test files on a shared scoped game-world fixture (no leaked worlds between tests), a versioned save system that migrates old saves and refuses newer ones, and Python editor and playtest tooling.

Shift — playable shop & delivery game — Aug – Sep 2026

Independent · Phaser 3, TypeScript, Vite · Live: xylardark.github.io/ShiftGame · Repo: github.com/XylarDark/ShiftGame

- Shipped a complete browser/PWA game loop with a fixed 60 Hz simulation and render interpolation, cached A* routing with auto-drive, and a tiered render budget that keeps it smooth on phones.

Portfolio Studio — Sep 2026 – present

Independent · React, TypeScript, Supabase · xylardark.github.io/Portfolio-Studio

- Shipped a public media-résumé site with an editor, embeddable work cards, and per-employer pages.

RELEVANT EXPERIENCE

Game Developer · OutFusion · Hamilton, ON — Feb 2022 – Dec 2022

- Implemented game UI/UX from the design docs; integrated and tested UI art across mobile and desktop with iterative playtesting; built and tested C# REST API integrations for player data.

Software Developer · TruckSail · Ancaster, ON — Feb 2020 – Jan 2021

- Wrote C++ firmware for truck ECU devices (STM32, CAN bus) with defect finding for reliability; built JS/Bootstrap UIs and optimized SQL; maintained GitHub workflows.

Software Developer & QA Tester · Evenica · Burlington, ON — 2018 – Sep 2019

- Built and tested client software (Andrew Peller, Young's Market Company, MM Realty International): T-SQL, C#, Linux, and Laravel (MySQL, PHP, JavaScript/jQuery, Vue.js).

Delivery Driver (Uber, retail delivery) · Hamilton, ON — 2022 – 2026

EDUCATION

Diploma, Computer Systems Technology — Software Development · Mohawk College · Hamilton, ON · 2019

Remote · Eastern Time (UTC−5 / UTC−4) · Canada